

The Algebra Behind the Karatsuba Algorithm

These steps may seem like magic, so let's dive into the algebra that shows why they work. Let's use 1,357 for x and 2,468 for y as the integers we want to multiply. Let's also consider a new variable, n , for the number of digits in x or y . Since a is 13 and b is 57, we can calculate the original x as $10^{n/2} \times a + b$, which is $10^2 \times 13 + 57$ or $1,300 + 57$, or 1,357. Similarly, y is the same as $10^{n/2} \times c + d$.

This means that the product of $x \times y = (10^{n/2} \times a + b) \times (10^{n/2} \times c + d)$. Doing a bit of algebra, we can rewrite this equation as $x \times y = 10n \times ac + 10^{n/2} \times (ad + bc) + bd$. With our example numbers, this means $1,357 \times 2,468 = 10,000 \times (13 \times 24) + 100 \times (13 \times 68 + 57 \times 24) + (57 \times 68)$. Both sides of this equation evaluate to 3,349,076.

We've broken the multiplication of xy into the multiplications of ac , ad , bc , and bd . This forms the basis of our recursive function: we've defined the multiplication of x and y by using multiplication of smaller numbers (remember, a , b , c , and d are half the digits of x or y) that approach the base case of multiplying single-digit numbers. And we can perform single-digit multiplication with a lookup table rather than multiplying.

So we need to recursively compute ac (the first step of the Karatsuba algorithm) and bd (the second step). We also need to calculate $(a + b)(c + d)$ for the third step, which we can rewrite as $ac + ad + bc + bd$. We already have ac and bd from the first two steps, so subtracting those gives us $ad + bc$. This means we need only one multiplication (and one recursive call) to calculate $(a + b)(c + d)$ instead of two to calculate $ad + bc$. And $ad + bc$ is needed for the $10^{n/2} \times (ad + bc)$ part of our original equation.

Multiplying by the $10n$ and $10^{n/2}$ powers of 10 can be done by padding zero digits: for example, $10,000 \times 123$ is 1,230,000. So, there's no need to make recursive calls for those multiplications. In the end, multiplying $x \times y$ can be broken into multiplying three smaller products with three recursive calls: `karatsuba(a, c)`, `karatsuba(b, d)`, and `karatsuba((a + b), (c + d))`.

With some careful study of this section, you can understand the algebra behind the Karatsuba algorithm. What I can't understand is how Anatoly